Representing Control in the World: Retained History, Physical Sources and Renewal in Q-RAIF

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October 2026

Abstract

What does a physical coupling actually represent? For a system coupled to a discarded port, the reduced map is completely positive but need not be multiplicative. We determine its exact algebra for finite-spectrum dephasing: a non-atomic port leaves precisely the coupling operator's commutant, which can be noncommutative. Retaining an oscillator permits exact closed-loop gates; retaining translational degrees of freedom gives a faithful representation in the closed propagation carrier. A single quadratic shell supplies the spatial Clifford relations, while evolution time and the mass anticommutator enter separately in the Lorentzian symbol.

A centered-input charging condition connects these structures without identifying their quadratic forms. Under an actual calibrated alignment of control and propagation generators, the work form is a positive pullback of the spatial shell. The same encounter can satisfy that all-state charge condition while its reduced exact algebra remains only a commutant. Neither persistence nor a positive cost alone selects a spacetime representation.

The physical realization retains the records and resources that the algebraic statement omits. A positive Gaussian material Hamiltonian supports acquisition, record-controlled force, sensing and another action, with bounds on physical energy moments and smooth interacting sources. A paid Gibbs bank renews a specified receiver, and one finite cursor approximates a sequence while retaining the complete old history. Full-field ring renewal, neutral compression and thermal sorting, local massive-body gates and a spatial work clock provide further finite implementations in their stated layouts. A common action includes clock and support reaction. Matching these quantum layouts to one covariant supported apparatus, sustained thermal renewal and jointly sharp exhaust remains a separate construction.

Keywords: Q-RAIF, physical representation, multiplicative domain, propagation symbol, quantum records, material response, thermal renewal

Technical appendices and their original numbering are retained in Q-RAIF C technical supplement (`QRAIF_C_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 Introduction

A control algebra specifies how operations compose. A physical interface must also say which operations its coupling transports exactly, what happens to the degrees of freedom carrying its records, and what resources allow another operation. An assumed homomorphism answers the first algebraic question by definition; it does not construct the interface.

Q-RAIF A identifies a Lorentzian read-out under propagation-symbol conditions, Q-RAIF B derives actuation and work from a field coupling, and the mass–energy paper relates spectral mass to finite read-outs [25, 26, 27]. Here we put the representation question between these structures and their physical implementation. The distinction is whether the relevant degrees of freedom are discarded or retained, not whether the environment has a large Hilbert space.

Exact transport. Sections 2–4 begin with the algebra supplied by a coupling. Its reduced Heisenberg map has an exact algebra, the multiplicative domain [29, 30, 31]. Theorem 4 computes it for a dephasing interaction. With a non-atomic discarded port it is the commutant of the coupling observable, including its full degeneracy blocks. The field-coupled flip then relates its remaining defect to drive energy and escaped information. These are architecture-specific statements, not a universal conservation-law prohibition on control [34, 35, 36].

Propagation and charging. When translations remain in the closed carrier, the internal algebra embeds faithfully. Section 5 distinguishes that exact embedding from the quadratic-space structure of a first-order propagation symbol. The latter requires a single shell, and its Lorentzian extension uses the supplied time and a mass anticommutator. Section 6 brings in Q-RAIF B's conditional work bridge: one centered independent field input charges every unknown carrier state equally precisely when each control generator combination has a scalar square. An actual calibration between these generators and the propagation symbol then relates the positive work form to the spatial shell. It does not turn that form into an indefinite spacetime metric. We show why the same calibrated encounter still need not represent the full control algebra after the port is discarded.

Retention and physical return. Section 7 gives the two decisive controls on the discarded-port argument: a genuinely fresh lost port leaves an irreversible defect, whereas a retained oscillator closes exact Weyl loops. The continuum loop and its finite adaptive error budget are developed in Appendix B. The retained case requires coherent routing, quadratures and a clock; an abstract embedding supplies none of those permissions. These are two realizations, not an exhaustive classification of physical interfaces.

The material construction in Sections 8–10 then follows one complete Gaussian Hamiltonian through acquisition, force, sensing, receiver replacement and another action. Records remain operators in their old joint history. Physical energy moments and smooth-source forms control unbounded field readings, rather than inferring stress convergence from trace distance alone. Receiver replacement uses a prepared Gibbs bank and transfers the spent correlations into its ledger.

The longer certificates are grouped by the physical interface they establish. Appendix A constructs the charged regulator, a finite mode with its remainder and the same-Hamiltonian source bounds. Appendix C treats the complete ring body, neutral gates, packet loading and thermal sorting. Appendix D treats the spatial work field and the separate massive-body gate. The remaining appendices retain the battery, outlets, thermal preparations, contact recoil, actual diagnostic return and current-stress compiler. Their distinct carriers are not identified by a common vocabulary. Section 11 gives a common action with clock/support reaction and states the matching still required.

Three resource limits matter. Bounded-energy receiver renewal takes longer and uses more clock states; fixed-duration finite rounds use growing clock energy; a spatial work clock uses horizon-dependent inertia and a supplied material frame. The ring converter's complete-output thermal deficit can approach zero in a prescribed finite layout. These results do not yet form one indefinitely renewed, fixed-capacity, covariant apparatus.

Unless stated otherwise, $\hbar = c = k_B = 1$. Energies expressed as frequencies acquire a factor $\hbar$ when units are restored, times and pulse areas use the corresponding frequency units, and entropy is in nats. The combinations $b + b^\dagger$, $i(b^\dagger - b)$ and occupation numbers are dimensionless; material smearings use the normalizations specified with their Hamiltonians below. All joint-state assertions include the stated old references; a preserved nonselective marginal need not remain unchanged after conditioning on a new record.

2 The algebraic statement

Let (V, q) be a finite-dimensional real vector space with a non-degenerate quadratic form and $Cl(V, q)$ the real Clifford algebra generated by V with $v^2 = q(v)\mathbf{1}$. When generators are to act as operators, those with $q(e) > 0$ are taken Hermitian and those with $q(e) < 0$ anti-Hermitian; this fixes an involution with $e^* = \operatorname{sign}(q(e))\, e$ on an orthogonal basis.

Definition 1 (Representation input)**.** A *representation* of a control algebra $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ in an environmental C^*-algebra $\mathcal{A}_{\text{ext}}$ is a unital $*$-homomorphism $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$.

Lemma 2 (Semisimplicity)**.** *$Cl(V, q)$ is semisimple. If $\dim V$ is even it is simple, and every nonzero algebra homomorphism out of it is injective.*

Proof. By the classification of real Clifford algebras [40, 41], $Cl(V, q)$ is a matrix algebra over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, or a direct sum of two such algebras, according to signature and dimension modulo eight. For even $\dim V$ it is a single matrix algebra. The kernel of a homomorphism is a two-sided ideal, and a simple algebra has none besides 0 and itself. □

Proposition 3 (Image closure)**.** *Let ϕ be as in Definition 1. Then $\operatorname{Im}\phi$ is a $*$-subalgebra of $\mathcal{A}_{\text{ext}}$ in which the Clifford relations hold. If $Cl(V, q)$ is simple, in particular when $\dim V$ is even, ϕ is injective. If it is the sum of two simple algebras, its image is isomorphic to the full algebra or to one of the two summands.*

Proof. Images of $*$-homomorphisms are $*$-subalgebras and relations descend. Simplicity gives injectivity. The only proper ideals of a sum of two simple algebras are zero and either summand; a unital representation has a proper kernel. □

Non-degeneracy is needed: for $q(x,y)=x^2$ on $\mathbb{R}^2$, the map $e_1 \mapsto \mathrm{diag}(1,-1)$, $e_2 \mapsto 0$ is a nonzero unital homomorphism into $M_2(\mathbb{R})$ that is not injective. An algebra homomorphism also does not compare signatures. That requires a quadratic-space morphism $j : (V,q) \to (W,Q)$ with $Q(jv) = q(v)$, and, where relevant, preservation of the involution and grading by the induced map of Clifford algebras. Proposition 3 provides neither the map ϕ nor such a j. Sections 3–5 ask where they can come from.

3 Couplings give channels

Let a finite-dimensional system S interact with an environment $\mathcal{E}$ through a unitary U on $\mathcal{H}_S \otimes \mathcal{H}_\mathcal{E}$, with the environment in a normal state σ. The induced Heisenberg map on system observables is

$$\Phi^*(A) = \mathrm{tr}_\mathcal{E}\big[(\mathbf{1}\otimes\sigma)\,U^*(A\otimes\mathbf{1})U\big]. \tag{1}$$

It is unital and completely positive, and by Stinespring's theorem every such map arises in this way [29]. It is in general not multiplicative. Its *multiplicative domain*

$$\mathcal{M}_\Phi = \{A :\ \Phi^*(A^*A) = \Phi^*(A)^*\Phi^*(A),\ \Phi^*(AA^*) = \Phi^*(A)\Phi^*(A)^*\} \tag{2}$$

is a C^*-subalgebra on which Φ^* is a $*$-homomorphism, and $\Phi^*(AB) = \Phi^*(A)\Phi^*(B)$ whenever $A \in \mathcal{M}_\Phi$ [30, 31]. It is the largest part of the system's algebra that the coupling represents exactly. For other elements we measure the failure by the *defect*

$$\delta(A) = \Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A) \ \geq 0. \tag{3}$$

Theorem 4 (Exactly represented algebra of a dephasing coupling)**.** *Let $X = \sum_x x P_x$ have finite spectrum, let E be self-adjoint on $\mathcal{H}_\mathcal{E}$, and $U = \exp(-iX\otimes E)$. Write $\chi(t) = \mathrm{tr}(\sigma e^{-itE})$ and $A_{xx'} = P_x A P_{x'}$. Then*

$$\Phi^*(A) = \sum_{x,x'} \chi(x'-x)\, A_{xx'}, \tag{4}$$

the relation $x \sim x'$ defined by $|\chi(x'-x)| = 1$ is an equivalence relation on the spectrum of X, and

$$\mathcal{M}_\Phi = \{A :\ A_{xx'} = 0 \text{ whenever } |\chi(x'-x)| < 1\}. \tag{5}$$

On $\mathcal{M}_\Phi$, Φ^* is an inner automorphism $A \mapsto W^*AW$ with $W = \sum_x u_x P_x$ for phases u_x. If $|\chi(t)| < 1$ for every nonzero difference t of eigenvalues of X, then $\mathcal{M}_\Phi = \{X\}'$ and Φ^* is the identity on it.

Proof. $U = \sum_x P_x \otimes e^{-ixE}$, so $U^*(A\otimes\mathbf{1})U = \sum_{x,x'} A_{xx'} \otimes e^{ixE}e^{-ix'E}$, and taking the expectation in σ gives Eq. (4).

If $|\,\mathrm{tr}(\sigma e^{-itE})| = 1$, write $\sigma = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; each $|\langle\psi_i|e^{-itE}|\psi_i\rangle| \leq 1$, so equality forces every ψ_i to be an eigenvector of e^{-itE} with one common eigenvalue $\chi(t)$. Hence e^{-itE} acts as the scalar $\chi(t)$ on the support of σ. If this holds for t_1 and t_2, it holds for $t_1 + t_2$ with $\chi(t_1+t_2) = \chi(t_1)\chi(t_2)$. Together with $\chi(0) = 1$ and $\chi(-t) = \overline{\chi(t)}$ this shows that $\sim$ is an equivalence relation, and that within a class with base point x_0 the phases $u_x = \chi(x - x_0)$ satisfy $\chi(x'-x) = \overline{u_x}u_{x'}$.

The diagonal blocks of the defect are

$$P_y\big[\Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A)\big]P_y = \sum_x \big(1 - |\chi(y-x)|^2\big)A^*_{xy}A_{xy}, \tag{6}$$

a sum of positive operators with nonnegative coefficients. It vanishes for all y only if $A_{xy} = 0$ whenever $|\chi(y-x)| < 1$; the condition with AA^* is the same with the roles of rows and columns exchanged. Conversely, if A has this block pattern, Eq. (4) gives $\Phi^*(A) = \sum \overline{u_x}u_{x'}A_{xx'} = W^*AW$, and these A form a $*$-algebra on which Φ^* is multiplicative. If no nonzero difference has $|\chi| = 1$, the classes are single eigenvalues, the algebra is $\bigoplus_x P_xB(\mathcal{H}_S)P_x = \{X\}'$, and W is scalar on each block. □

Corollary 5 (Continuous spectrum forbids exact coupled control)**.** *If the spectral distribution of E in σ has no atoms, then $|\chi(t)| < 1$ for every $t \neq 0$, so $\mathcal{M}_\Phi = \{X\}'$. In particular this holds for every normal state when E has purely continuous spectrum. Every exactly represented element A commutes with X, and hence with the interaction on its natural domain. No nonzero rescaling of the coupling changes this conclusion.*

Proof. The equality $|\int e^{-ite}\,d\mu(e)| = 1$ requires the phase e^{-ite} to be constant μ-almost everywhere. For $t \neq 0$ its level sets are translates of the countable lattice $(2\pi/t)\mathbb{Z}$, and a non-atomic probability measure gives each such set measure zero. Absolute continuity is sufficient but unnecessary. □

A field quadrature, or the smeared current Q_g of Paper B, has purely absolutely continuous spectrum, so Corollary 5 applies to every normal drive state, whatever its energy. For other couplings, $|\chi(t)| = 1$ requires support in an eigenspace of e^{-itE}, not necessarily an eigenstate of E. Periodic spectra can give exact revivals: with $X = \pi Z$, $E = \mathrm{diag}(0,1)$ and $\sigma = |+\rangle\langle+|$, the interaction unitary is $I \otimes \mathrm{diag}(1,-1)$ and the system channel is identity, although σ is not an E eigenstate. A Fock state in a number-type coupling gives another finite-energy possibility. The quadrature obstruction concerns the specified single coupling; extending it to composite local controls requires a separate argument.

4 The field-coupled flip

Paper B couples a degenerate bit to one mode of a free chiral field through $U = \exp(-i\lambda XQ)$, $\lambda > 0$, with $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ and $[Q,P] = 2i$ [26]. Here $E = \lambda Q$ and the eigenvalues of X are ± 1. Write $\chi = \mathrm{tr}(\sigma e^{-2i\lambda Q})$ and $\sigma_\pm = e^{\mp i\lambda Q}\sigma e^{\pm i\lambda Q}$ for the field states conditioned on $X = \pm 1$. A flip of Z-diagonal states fails with probability $\epsilon = (1 + \mathrm{Re}\,\chi)/2$.

Theorem 6 (Representation defect of the field-coupled flip)**.** *Let the drive mode be in a normal state σ with mean occupation $N = \mathrm{tr}(\sigma b^\dagger b) < \infty$.*

1. *In the X eigenbasis, Φ^* multiplies the two off-diagonal entries by χ and $\overline{\chi}$, and $\mathcal{M}_\Phi = \{X\}'$, the algebra generated by X.*

2. *For every A with vanishing diagonal in the X basis, in particular Z and Y, $\delta(A) = (1 - |\chi|^2)\,\mathrm{diag}(|A_{-+}|^2, |A_{+-}|^2)$; thus $\delta(Z) = \delta(Y) = (1 - |\chi|^2)\mathbf{1}$.*

3. $1-|\chi|^2 \;\geq\; \dfrac{\arccos^2(\operatorname{Re}\chi)\,|\chi|^2}{(2N+1)^2}$.

4. $1-|\chi|^2 \geq D(\sigma_+,\sigma_-)^2$*, with equality when* σ *is pure, where* D *is the trace distance.*

Proof. Item 1 is Theorem 4 with eigenvalue differences ± 2 and Corollary 5. Item 2 follows from Eq. (4) by multiplying out 2×2 matrices.

Item 3 uses Paper B's two quadrature bounds, valid for a normal (density-operator) drive state of finite occupation and $[Q,P]=2i$ [26]: with $G(c)=\arccos^2 c$,

$$\langle Q^2\rangle \geq \frac{G(\operatorname{Re}\chi)}{4\lambda^2}, \qquad (\Delta P)^2 \geq \frac{4\lambda^2|\chi|^2}{1-|\chi|^2}.$$

The first is spectral Jensen; the second is the unitary uncertainty bound. Since $4N+2=\langle Q^2+P^2\rangle$,

$$\frac{4\lambda^2|\chi|^2}{1-|\chi|^2} \leq \langle P^2\rangle = 4N+2-\langle Q^2\rangle \leq 4N+2-\frac{G(\operatorname{Re}\chi)}{4\lambda^2}.$$

With $u=4\lambda^2$, $a=G(\operatorname{Re}\chi)$ and $A=4N+2$, this reads $1-|\chi|^2 \geq u^2|\chi|^2/(Au-a)$ for some $u>a/A$. The right side is minimized over u at $u=2a/A$, where it equals $4a|\chi|^2/A^2$.

Item 4. If $\sigma=|\psi\rangle\langle\psi|$, the conditioned states are pure with overlap $\langle e^{i\lambda Q}\psi, e^{-i\lambda Q}\psi\rangle=\chi$, and $D^2=1-|\chi|^2$. For mixed σ, applying $e^{\mp i\lambda Q}\otimes\mathbf{1}$ to a purification of σ gives purifications of $\sigma_\pm$ with overlap χ, so Uhlmann's theorem gives fidelity $F(\sigma_+,\sigma_-)\geq|\chi|$, and $D\leq(1-F^2)^{1/2}$ [32, 33]. □

Corollary 7 (The energy–accuracy bound as a representation statement)**.** *For a flip with failure probability* $\epsilon\in(0,\frac{1}{2})$ *and* $\theta_\epsilon=\arccos(2\epsilon-1)$*,*

$$N \;\geq\; \frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}}-\frac{1}{2}. \tag{7}$$

Proof. $\operatorname{Re}\chi=2\epsilon-1$ gives $|\chi|^2\geq(1-2\epsilon)^2$, hence $1-|\chi|^2\leq 4\epsilon(1-\epsilon)$. Combining with item 3 of Theorem 6 gives $4\epsilon(1-\epsilon)\geq\theta_\epsilon^2(1-2\epsilon)^2/(2N+1)^2$. □

This is exactly the occupation bound of Paper B. Read in the present terms, an accurate flip requires the coupling to represent Z almost multiplicatively, and the defect $1-|\chi|^2$ cannot vanish at finite occupation. Item 4 identifies the defect with information: for pure drives it is the squared distinguishability of the two field states that record the value of X. The field is then a fragment in the sense of the HAFF paper on measurement, with conditional overlap $|\chi|$; that paper determines which pointer effects such a fragment can implement [28], and the same overlap fixes the representation defect here. The exactly represented algebra $\{X\}'$ is the pointer algebra of the decoherence literature [37]: the control interface transports exactly the part of the system that the environment records.

For the phase-fixed half-turn squeezed family, Paper B gives occupations 2.958, 7.330 and 24.329 at $\epsilon=.05,.01,.001$, against bound values 2.28, 6.74 and 23.80. Its separately optimized finite quadrature-grid values are variational minima, not certified global optima; the continuous supporting ground-state argument has its own assumptions. Mixtures of coherent states require occupation of order $\pi^2/(16\epsilon)$ [26].

4.1 Same clock as mode matching

The requirement that control and environment share a clock, so that their operational windows overlap, has a precise form in the field-coupled flip. The switch g defines a normalized mode function f and the mode operator $b_f = \int \overline{f(\omega)}\, a_\omega \, d\omega$ with $Q_g = \lambda(b_f + b_f^\dagger)$ [26]. A drive is prepared in some normalized mode f', by a source with its own clock.

Proposition 8 (Mode overlap)**.** *Let the drive occupy mode f' with mean occupation N', all modes orthogonal to f' being in the vacuum, and let $c = \int \overline{f(\omega)} f'(\omega)\, d\omega$. Then the bit channel equals the channel of Theorem 6 for the reduced state of mode f, which is the output of a pure-loss channel of transmissivity $|c|^2$ and has mean occupation $|c|^2 N'$. A flip with failure ϵ therefore needs*

$$N' \;\geq\; \frac{1}{|c|^2}\left[\frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}\right]. \tag{8}$$

If $f'(\omega) = f(\omega)e^{i\omega\Delta}$, a copy of f delayed by Δ, then $c = \int |f(\omega)|^2 e^{i\omega\Delta} d\omega$. For a positive-frequency field this integral is over $\omega > 0$. Extending a narrowband Gaussian spectral density of width Ω to the full real line gives $|c| \simeq e^{-\Omega^2\Delta^2/2}$ and an occupation penalty $e^{\Omega^2\Delta^2}$ in the range where this approximation controls the overlap. The exact positive-frequency integral, not this Gaussian approximation, determines the cost at long delays.

Proof. Write $f = \bar{c} f' + r$ with $\int \overline{f'}\, r\, d\omega = 0$ and $\|r\|^2 = 1 - |c|^2$. Since $g \mapsto b_g$ is antilinear, $b_f = c\, b_{f'} + b_r$, where $[b_r, b_r^\dagger] = 1 - |c|^2$ and b_r commutes with $b_{f'}$ and $b_{f'}^\dagger$. The mode $r/\|r\|$ is orthogonal to f' and in the vacuum, independent of mode f', so the state of mode f is that of a beam splitter of transmissivity $|c|^2$ with vacuum in the other port, and its occupation is $|c|^2 N'$. Q_g involves only mode f, so the bit channel depends only on this reduced state, and Corollary 7 applies to it. Translation gives the exact Fourier overlap. The Gaussian specialization follows only after the stated extension of the frequency support. □

The prepared drive has energy $N'\bar{\omega}'$, while only the fraction $|c|^2$ of its occupation reaches the selected mode. This is a charged preparation resource, not automatically dissipated heat. The source/switch clock offset therefore enters Paper B's feedback acceptance as a mode-matching loss.

5 A representation generated by propagation

The situation is different when the "environment" is part of the same closed dynamics. In the local propagation model of Paper A, a particle with internal space $\mathbb{C}^s$ moves on a lattice Λ by a unitary step $W = \sum_{h \in N} T_h \otimes A_h$, where T_h translates by h [25]. The internal algebra is represented by

$$\phi : M_s(\mathbb{C}) \to B\big(\ell^2(\Lambda) \otimes \mathbb{C}^s\big), \qquad \phi(A) = \mathbf{1} \otimes A, \tag{9}$$

a faithful unital $*$-homomorphism, and $[\phi(A), W] = \sum_h T_h \otimes [A, A_h]$ is nonzero unless A commutes with every A_h. The requirement of Definition 1 together with nontrivial coupling is met exactly, because nothing is traced out: the translational degrees of freedom and the internal ones evolve under one unitary. The contrast with Section 4 is the content of

Corollary 5: discarding a non-atomic port restricts exact multiplicativity to the coupling commutant.

The propagation also supplies the quadratic-space morphism that Section 2 found missing. Near a band point the step has a first-order symbol $H(q) = \sum_i q_i\alpha_i + \mu\beta$ with Hermitian coefficients [25].

Proposition 9 (The propagation symbol as a quadratic-space morphism)**.** *Let $j(q) = \sum_i q_i\alpha_i$ with $\alpha_1, \dots, \alpha_d$ linearly independent. Then $j(q)^2 = Q(q)\mathbf{1}$ for a quadratic form Q if and only if, for $\mu = 0$, every eigenvalue of $H(q)$ lies on one shell $\omega^2 = Q(q)$. In that case Q is positive definite, j is an isometry from $(\mathbb{R}^d, Q)$ into* $\mathrm{Herm}(s)$ *with the normalized Hilbert–Schmidt form, and j extends to a unital $*$-homomorphism $Cl(\mathbb{R}^d, Q) \to M_s(\mathbb{C})$. If moreover $\mu \neq 0$, $\beta^2 = \mathbf{1}$ and $\{\alpha_i, \beta\} = 0$, then*

$$J(\omega, q) = \omega\beta - \sum_i q_i\beta\alpha_i \tag{10}$$

satisfies $J(\omega, q)^2 = (\omega^2 - Q(q))\mathbf{1}$ and extends to a unital homomorphism $Cl(\mathbb{R}^{1+d}, \eta) \to M_s(\mathbb{C})$ with $\eta(\omega, q) = \omega^2 - Q(q)$, which preserves the involution in which β is Hermitian and the $\beta\alpha_i$ are anti-Hermitian.

Proof. The first statement is the single-shell theorem of Paper A [25] with $\mu = 0$; isometry follows from $\frac{1}{s}\operatorname{tr} j(q)^2 = Q(q)$, and the extension from the universal property of Clifford algebras. For J, expand J^2 using $\beta^2 = \mathbf{1}$ and $\beta\alpha_i\beta = -\alpha_i$: the cross terms $\omega q_i\{\beta, \beta\alpha_i\}$ vanish and $\sum q_i q_j \frac{1}{2}\{\beta\alpha_i, \beta\alpha_j\} = -Q(q)$. Hermiticity of β and anti-Hermiticity of $\beta\alpha_i = -\alpha_i\beta$ follow from $(\beta\alpha_i)^* = \alpha_i\beta$. □

J is the map $j : (V, q) \to (W, Q)$ of Section 2 with $(V, q) = (\mathbb{R}^{1+d}, \eta)$, a Lorentzian quadratic space, and W the real span of β and the $\beta\alpha_i$ in $M_s(\mathbb{C})$, with the form defined by $Y^2 = Q_W(Y)\mathbf{1}$. The spatial scalar-square condition is equivalent to the specified single-shell property, so no internal band has a different squared dispersion. The Lorentzian extension additionally uses the time parameter and the mass anticommutator. A massless one-dimensional carrier can have only one chiral branch; a full two-sided cone requires both frequency signs, as Q-RAIF A explains [25].

The canonical inclusion ϕ remains exact on the lattice. It is the linear quadratic-space description of its propagation symbol that generally holds only at leading order. For $s = 2$ and $W(k) \in SU(2)$, the effective generator $H_{\mathrm{eff}}(k) = (i/\tau)\operatorname{Log} W(k)$ is traceless, so $H_{\mathrm{eff}}(k)^2 = \omega(k)^2\mathbf{1}$ holds at every k; what fails is linearity and isotropy of $k \mapsto H_{\mathrm{eff}}(k)$. In the Weyl automaton of Paper A, $H_{\mathrm{eff}}(k) = c\,\sigma \cdot k$ up to corrections of relative order $|k|\ell$, which include a parity-odd anisotropic term. In the split-step walk the quadratic identity $\omega^2 = c_{\mathrm{eff}}^2 q^2 + \mu^2$ fails at order $(qa)^4$, with coefficient $-\theta^2/45 + O(\theta^4)$ in lattice units [27]. These orders are the precise sense in which a lattice propagation gives an approximate representation.

6 Calibrated work and exact access

The propagation symbol and an actuator can use the same internal generators without being the same physical system. Q-RAIF B supplies a specific way to test their alignment [26].

Let $A(a) = \sum_i a_i A_i$ be a fixed real linear family of dimensionless Hermitian operators. During an encounter the command a is held fixed, the carrier free Hamiltonian is zero or commutes with $A(a)$, and a nonzero real smooth compact switch g couples it to one chiral current as $g(t)A(a)j(t)$. The field input is independent, centered and of finite energy. With $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, the same encounter has

$$W_\rho(a) = \kappa_g \operatorname{tr}[\rho A(a)^2]. \tag{11}$$

This is field energy delivered by switching, not a total-device heat charge. A nonzero commuting carrier Hamiltonian has no energy change during this encounter; cancelling a non-commuting drift is an additional operation.

The charge is independent of every carrier state ρ for every command precisely when $A(a)^2 = Q_{\mathrm{ctrl}}(a)I$. Polarization gives $\{A_i, A_j\} = 2(Q_{\mathrm{ctrl}})_{ij} I$. For an injective family Q_{ctrl} is positive definite. The implication is from an all-state charging requirement, not from persistence, reachability or positivity of an average cost. The field shift is $j(u) \mapsto j(u) + A(a)g'(u)/2$. The second Magnus term can be a system operation $e^{i\vartheta A(a)^2}$ rather than a global phase; it commutes with the field and does not change this work calculation [26].

To relate it to propagation, write the spatial symbol as $j_{\mathrm{prop}}(p) = \sum_i p_i \alpha_i$, with energy units and $j_{\mathrm{prop}}(p)^2 = Q_{\mathrm{prop}}(p)I$. Let C be the measured map from commands to momenta and $E_* > 0$ an energy scale. The actual coupling must satisfy

$$A(a) = j_{\mathrm{prop}}(Ca)/E_*.$$

The equality specifies a physical alignment. It cannot be inferred by renaming two positive forms.

Corollary 10 (Aligned charging and the reduced exact algebra)**.** *Under these encounter and alignment assumptions,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2} C^T Q_{\mathrm{prop}} C. \tag{12}$$

For a normal field input and a discarded field port, the encounter's multiplicative domain is nevertheless

$$\mathcal{M}_a = \{j_{\mathrm{prop}}(Ca)\}'. \tag{13}$$

If C is invertible, the common exact algebra for all commands is $\bigcap_a \mathcal{M}_a = \{\alpha_1, \ldots, \alpha_d\}'$. For an irreducible complex matrix representation this common algebra is scalar.

Proof. The scalar-square identity inserted in Eq. (11) gives Eq. (12), the aligned-work result of Q-RAIF B. The nonzero switched current is a field quadrature with non-atomic spectral distribution in every normal input. Corollary 5 applies with $X = A(a)$ and $E = Q_g$. The commuting Magnus factor is scalar for this aligned family, so it does not change the domain. Commuting free carrier evolution preserves the same block algebra. Commuting with every linear combination is equivalent to commuting with each generator when C is invertible. The last assertion is the matrix commutant criterion for irreducibility. □

The corollary combines the charging result of Q-RAIF B with the multiplicative-domain calculation; it is not a further selection principle for Clifford algebras. For one command the domain retains full degenerate eigenspace blocks and can be noncommutative. Over all commands it can become much smaller. Exact closed-carrier representation, state-independent work and exact reduced control are therefore different properties.

Two failures keep the calibration operational. For $A(a) = a_1 Z + a_2 I$, the maximally mixed state costs $\kappa_g(a_1^2 + a_2^2)$, but the two Z eigenstates cost $\kappa_g(a_1 \pm a_2)^2$; one average-state test does not establish the all-state condition. For $j_{\mathrm{prop}}(p) = p_1 X + p_2 Z$ and $C = \mathrm{diag}(1, 2)$ in units $E_* = 1$, the unchanged Euclidean propagation shell gives $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$. Without C, work does not identify the propagation form.

The time direction and mass anticommutator remain the additional conditions of Proposition 9. In $\omega^2 - Q_{\mathrm{prop}}(p)$, ω is conjugate to the supplied evolution time. An independent, noncentered drive or a correlated input has a stress cross term $g'(u)\langle A(a)j(u)\rangle$; a coherent gate cannot inherit the centered calibration price for all states. Nor does the calibration construct a shared propagation medium, pause its noncommuting drift for free, or maintain its controls indefinitely.

7 Discarded ports and coherent return

A discarded record port and a retained environment permit different controls. Write $\chi(t) = \mathrm{Tr}(\sigma_E e^{-itE})$ for the port's spectral characteristic function. Section 3 derives the single-coupling channel and its multiplicative domain; the following bound survives arbitrary later controls that cannot reach the lost port.

Proposition 11 (A fresh port leaves an irreversible defect)**.** *Suppose a finite-dimensional system first undergoes ideal unitary controls and then one coupling $e^{-iX\otimes E}$ to an initially independent port, with X having at least two distinct eigenvalues and the environment's spectral distribution non-atomic. Let F collect every auxiliary port accessible later; just before this coupling the joint state is $\rho_S \otimes \sigma_E \otimes \tau_F$ for every system input. If E is thereafter inaccessible, no subsequent system channel can make the complete channel unitary on the full system algebra. More precisely, let Λ include all the preceding and subsequent controls, and let $\mathcal{V}$ be any unitary channel. For every pair of distinct eigenvalues x, x',*

$$\frac{1}{2}\|\Lambda - \mathcal{V}\|_\diamond \;\geq\; \frac{1 - |\chi(x' - x)|}{2}. \tag{14}$$

Proof. Choose eigenvectors $|x\rangle, |x'\rangle$ and the orthogonal states $|\pm\rangle = (|x\rangle \pm |x'\rangle)/\sqrt{2}$. Their trace distance is one before the coupling and $|\chi(x' - x)| < 1$ afterwards. All subsequent operations induce a system channel and cannot increase this distance. Pull the two inputs back through the preceding unitaries and call them $\rho_\pm$. If $e_\pm = D(\Lambda\rho_\pm, \mathcal{V}\rho_\pm)$, the triangle inequality gives $1 \leq e_+ + |\chi(x' - x)| + e_-$. Thus $\max e_\pm \geq (1 - |\chi|)/2$, which also bounds half the diamond norm from below. □

The independence concerns the entire later-accessible auxiliary system jointly, including any purification of the discarded port. A later register correlated with that port can reveal which phase was applied and allow its reversal; the system's reduced channel alone then

does not describe the available recovery operation. For a qubit and no later noise, the bound is attained: remove the phase of χ, and the remaining dephasing is the mixture $(1+|\chi|)\mathcal{I}/2 + (1-|\chi|)\mathcal{Z}/2$. Its half diamond distance from $\mathcal{I}$ is $(1-|\chi|)/2$.

Proposition 12 (An exact interface from a retained oscillator)**.** *Let $Q = b+b^\dagger$, $P = i(b^\dagger - b)$ satisfy the Weyl relations, so $[Q,P] = 2i$, and let X be Hermitian with finite spectrum. Then*

$$e^{-i\lambda XQ}e^{-i\mu XP}e^{i\lambda XQ}e^{i\mu XP} = e^{-2i\lambda\mu X^2} \otimes I. \tag{15}$$

Every initial oscillator state is returned exactly. The induced system map has multiplicative domain $M_s(\mathbb{C})$, including operators that do not commute with the coupling observable X.

Proof. In an X eigenblock, the two pulse generators have scalar commutator

$$[-i\lambda xQ, -i\mu xP] = -2i\lambda\mu x^2.$$

The Weyl group commutator is therefore that scalar phase. Summing the spectral blocks gives (15). □

For a qubit choose $X = (I+\sigma_x)/2$. With $\lambda\mu = \pi/2$, the loop is $e^{-i\pi X} = -\sigma_x$: an exact corrective flip, even with a vacuum or thermal oscillator input. The intermediate system–oscillator correlations are returned, not discarded. Conditional displacement loops of this kind are established geometric gates [24]; the point here is their implication for the interface. Continuous spectrum alone cannot prohibit exact control of a retained port.

Corollary 13 (One retained oscillator gives universal qubit control)**.** *Suppose the four pulses of Proposition 12 can be applied through either $P_z = (I+\sigma_z)/2$ or $P_x = (I+\sigma_x)/2$, with arbitrary real pulse areas. Every qubit unitary, up to global phase, is a sequence of at most three closed loops, and the oscillator returns after each loop.*

Proof. A loop with $2\lambda\mu = \alpha$ implements $e^{-i\alpha P_z} = e^{-i\alpha/2}e^{-i\alpha\sigma_z/2}$, and similarly for P_x. The z–x–z Euler decomposition of a qubit unitary therefore gives three loops. For example the three angles $\pi/2, \pi/2, \pi/2$ give a Hadamard up to global phase. Each identity holds on the full system–oscillator tensor product, so composition returns even an oscillator initially correlated with another system. □

For a larger system the same construction supplies the one-parameter groups e^{-itP_j}. Full unitary control follows when the real Lie algebra generated by iP_j is $\mathfrak{u}(s)$; generation of the associative matrix algebra alone is not the required controllability condition. The pulse permissions, including the choice between noncommuting projectors, remain physical inputs.

The loop is not costless. For a centred input and a mode of frequency ω, the largest conditional occupation increase along the four corners is $(\lambda^2+\mu^2)x^2$. Minimizing it for the qubit flip gives $\lambda = \mu = \sqrt{\pi/2}$ and peak mean stored energy $\pi\omega\langle X\rangle \le \pi\omega$. The mode's final energy is unchanged; an ideal reversible switch can recover the intermediate work. This is a stored-energy statement, not a total-device dissipation estimate. It assumes access to both quadratures, their controlled pulse order, and coherent retention until closure. A travelling field that has left the apparatus need not supply these permissions. An implementation with smooth local spacetime couplings must establish them separately.

Thus a record carried away by an inaccessible port constrains future control, while a coherently returned record can disappear without thermal erasure. Retaining an interface can replace a preparation-accuracy requirement by a closure requirement. The oscillator, its routing, clock and switching controls are part of that retained interface; they are not supplied by an abstract algebra homomorphism.

8 What must be retained by a material interface

The representation results answer an algebraic question about a specified interaction. A material interface must additionally let an acquired record change a later operation on the same body. The source then belongs to that joint dynamics; it is not assigned from the program's name.

For the next construction, fix a finite Dirichlet interval and retain all modes of the positive Gaussian operator K built in Appendix A.1. Its energy form contains the complete carrier square $\|\phi' + a\Theta\|^2$, the material gradient and a strictly positive material mass matrix. The interval gives a spectral gap. The field Hamiltonian is $H_F = d\Gamma(\sqrt{K})$ with one fixed vacuum subtraction. Projecting onto the bound material mode does not remove the orthogonal modes: their reaction is explicitly bounded in Proposition 23. The charged regulator in that appendix is a separate finite-resolution construction, not a proof that this static Gaussian realization equals the nonlinear charged continuum.

The following section keeps this complete K, the finite controller, its records and all old references. Completing the controlled field squares gives a positive Hamiltonian and a physical rms bound. The operator retarded equation makes acquisition affect the force, the actual field affects another sensor, and its new record can control the next force. The smooth interacting-source estimate retains both cross terms and connected fluctuations.

Receiver renewal has a narrower but attainable target. A prepared matched-gap Gibbs cell can replace one force-idle receiver, moving its old correlations into the spent bank. The material and the bank's purifier remain present. Section 10 gives this certificate and compiles a specified finite sequence on one cursor. Renewing the entire interacting equilibrium while preserving arbitrary held quantum content is a different task, with a product-extension obstruction.

The longer implementations establish additional permissions: full-ring reset and spectral sorting (Appendix C), a spatial work field and a separate massive-body gate (Appendix D), a guarded battery (Appendix E), local export (Appendix F), thermal preparation (Appendix G), and contact reaction/recoil (Appendix H). Their lattice, ring, copropagating and bidirectional carriers have different supports and Hamiltonians. Their estimates must be matched before they describe one apparatus.

9 Acquired quantum records on the complete material

The force calculation in Appendix A.2 conserves a classical record during its drive. An acquired record need not be classical in the full old history. Keep the complete finite-interval K and $H_F = d\Gamma(\omega)$, $\omega = \sqrt{K}$, of Section A.3. Let the finite controller include probes,

record cells and program cursor. Its nonnegative Hamiltonian H_D need not commute with the force amplitudes. The following construction therefore includes acquisition and a later record-controlled response within one autonomous dynamics.

9.1 A positive common Hamiltonian and a uniform physical moment

Let B_a be finitely many commuting bounded self-adjoint controller operators and h_a real smooth compact material smearings. Define

$$F = \sum_a B_a h_a, \qquad C_{\min} = \frac{1}{2}\sum_{ab} B_a B_b \langle h_a, K^{-1} h_b\rangle,$$
$$H_{\rm tot} = H_D + H_F - \sum_a B_a Q(h_a) + C_{\min}. \tag{16}$$

The full canonical material fields obey $[Q(h), P(k)] = i\langle h, k\rangle$, with $Q(h) = [a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)]/\sqrt{2}$. This is the canonical normalization of the material Hamiltonian, distinct from the earlier dimensionless $[b + b^\dagger, i(b^\dagger - b)] = 2i$ convention. The amplitudes commute at equal Schrödinger time, not necessarily at different Heisenberg times. In particular a cursor-controlled record force and a probe force can commute while acquisition edges in H_D change those same records.

Proposition 14 (Positive quantum interface and physical rms bound)**.** *The Hamiltonian* (16) *is self-adjoint and nonnegative. Its domain is equivalent to* $\operatorname{Dom} H_F$*. For a normalized initial vector* ψ *in that domain, including all old references, put*

$$L_F = \sqrt{2}\sum_a \|B_a\|\|\omega^{-1}h_a\|, \qquad c_F = \frac{1}{\sqrt{2}}\sum_a \|B_a\|\|\omega^{-1/2}h_a\|,$$
$$A_F = \|H_{\rm tot}\psi\| + \|H_D\| + \|C_{\min}\| + c_F, \qquad M_F = \left[\frac{L_F + \sqrt{L_F^2 + 4A_F}}{2}\right]^2.$$

Then $\|H_F e^{-itH_{\rm tot}}\psi\| \le M_F$ *for every* t*. If the initial cursor lies in a force-idle region and* $\|H_D\|$*, coupling norms and initial physical rms stay fixed, this certificate does not grow with the finite program length.*

Proof. Resolve the common spectrum of the B_a into projections P_b and write $f_b = \sum_a b_a h_a$, $z_b = -\omega^{-3/2} f_b/\sqrt{2}$. The finite interval has a gap; hence the finite controlled Weyl unitary $D = \sum_b P_b \otimes W(z_b)$ exists and $D^* H_F D = H_F - \sum B_a Q(h_a) + C_{\min}$. It follows that $H_{\rm tot} = D^* H_F D + H_D \ge 0$. The linear field bound

$$\|Q(h)\zeta\| \le \sqrt{2}\|\omega^{-1}h\|\|H_F^{1/2}\zeta\| + \|\omega^{-1/2}h\|\|\zeta\|/\sqrt{2}$$

also makes the force infinitesimally H_F-bounded, giving the graph domain statement by the closed operator realization. For $\psi_t = e^{-itH_{\rm tot}}\psi$ and $x = \|H_F\psi_t\|$, conservation of $\|H_{\rm tot}\psi_t\|$ gives $x \le A_F + L_F\sqrt{x}$, since $\langle H_F\rangle_t \le x$. Solving this quadratic inequality proves the bound. Spectral-core approximation extends the calculation to the stated domain. In a force-idle initial region, $\|H_{\rm tot}\psi\| \le \|H_F\psi\| + \|H_D\|$. Every step tensors with the actual reference identity. □

The displacement D proves positivity; it is not a free preparation of a dressed vacuum. This finite-interval argument is not an infinite-line static-dressing theorem. A positive nearest-neighbor Feynman cursor with bounded edge gates can have $\|H_D\| \le 4J + \|H_{\mathrm{data}}\|$ independently of its site count [4]. This removes a program-length factor from the physical moment bound, not from cursor space, execution time or preparation cost. For the uniform-J cursor, Foundation III's lemma "Wholly idle input and a finite completion window" in the appendix "A Finite Arrival Bound at Fixed Hopping Energy" bounds arrival for a packet supported wholly in the input idle region of the pure gate-dressed hopping comparison [50]. Cell gaps and forces require the subsequent Duhamel control; this is neither the spin-clock nor the spatial work-clock estimate below. Linear finite-system/field Hamiltonians and their domain analysis are established tools [5]; the issue here is their common retained-record interface.

The field obeys the operator retarded equation

$$Q(t) = \cos(\omega t)Q(0) + \omega^{-1}\sin(\omega t)P(0) + \int_0^t \omega^{-1}\sin[\omega(t-s)]F(s)\,ds. \tag{17}$$

It follows by solving $\dot{Q} = P$, $\dot{P} = -KQ + F(t)$ as weak smeared identities on the energy core. The $B_a(s)$ in $F(s)$ are the actual joint Heisenberg operators. Their expectations are not an externally supplied force history. With material-only force, the exterior carrier formula of Section A.2 holds with the complete operator $\Delta\Theta$:

$$\Delta\phi_R(u) = -\tfrac{1}{2}\int a(y)\Delta\Theta(y, u+y)\,dy, \qquad \Delta\phi_L(v) = \tfrac{1}{2}\int a(y)\Delta\Theta(y, v-y)\,dy.$$

An additional carrier-smeared probe contributes its own retarded source. Quadratic stress uses correlations of these operators, not only the square of their mean.

9.2 Actual acquisition, sensing and the next action

For a probe P and a blank record R, the unitary $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$ acquires the probe's Y reading. Retain P, R and their old references. During a subsequent block that conserves R, the force of Section A.2 acts as $V = \sum_{r=0,1} |r\rangle\langle r| \otimes e^{ir^2\Phi_h} W(rz_h)$. For actual initial material/reference blocks $\sigma_{rr'}$, the output is

$$\sum_{rr'} |r\rangle\langle r'| \otimes e^{i(r^2 - r'^2)\Phi_h} W(rz_h)\sigma_{rr'}W(r'z_h)^*. \tag{18}$$

This is not a sampled classical force. For a pure complete input with both record branches present, dephasing R changes the full retained state by half trace distance $\sqrt{p(1-p)}$, where $p = \Pr(R=1)$. Indeed the deleted off-diagonal block connects two orthogonal vectors of norms $\sqrt{p}, \sqrt{1-p}$ and has two eigenvalues $\pm\sqrt{p(1-p)}$.

Now let a fresh sensor S start in $|+x\rangle$, couple by $e^{igZ_S Q(b)}$ to the material after its actual propagation, and acquire its Y value into a cell E. The latter starts with thermal occupation ν, independently prepared with its purifier retained. For a centered Gaussian

material initially independent of the first probe, let μ_r be the actual conditional mean of $Q(b)$ and V_b its conditional variance after the force. Then

$$\Pr(E = 1 \mid R = r) = \frac{1}{2} - \frac{1 - 2\nu}{2} e^{-2g^2 V_b} \sin(2g\mu_r). \tag{19}$$

The sensor's Y mean is $-\operatorname{Im}\langle e^{2igQ(b)}\rangle$; the Gaussian characteristic function and the mixed cell's visibility $1-2\nu$ give the formula. The acquisition acts on the same disturbed material, and the actually acquired n_E can control the next force. Its energy and source are part of that dynamics. For a correlated or non-Gaussian input use its actual characteristic function, not this Gaussian specialization.

Before sensing, a conserved source record and centered product input give $\langle \Delta j\rangle = p j_h$ but $\Delta\langle : j^2 : \rangle = p j_h^2$, where j_h is the complete driven carrier current. The connected contribution is $p(1-p)j_h^2$, not an extra energy charge. Similarly the full added material/carrier energy is pE_h, of which $p^2 E_h$ is the energy of the mean displacement. Record dephasing leaves this immediate field marginal unchanged while changing the old joint history. Equal mean source therefore does not establish equal later control.

The formulas describe the sequential block. Under the autonomous Hamiltonian (16), acquisition, force and sensing instead occupy cursor regions that a dispersive cursor can revisit. Its deadline statistics require the actual joint evolution, without postselection or assignment of the sequential probabilities.

A one-mode check retains the old probe reference and the second cell's purifier: $p = .2766071$, $\Pr(E = 1 \mid R = 1) = .4267590$, dephasing distance .4473205, and first-acquisition probe/reference change below 4.1×10^{-16}. Cutoffs $6, 10, 14$ reduce Gaussian-law error from 1.04×10^{-7} to 1.11×10^{-16}. Field energy .0418368 splits into mean .0115724 and connected .0302645; sensing adds $.04805 = g^2/2$ at $g = .31$. This checks the sequence, not the full spatial K theorem.

A separate 31-site positive cursor diagnostic executes actual first acquisition, record force, sensing, second acquisition, receiver/bank SWAP and a later force controlled by the spent cell's actual record. With both cell purifiers retained, its unpostselected deadline $t = 40$ has field energy .0300724, physical rms at most .186211 against the uniform bound 3.01330, and energy residual below 1.9×10^{-13}. But only .857524 of its cursor probability is after the SWAP region; its receiver occupation is .157712, exceeding the prepared bank value .13 by .027712. Unfinished and returning cursor branches retain pre-SWAP data and can revisit acquisition and force regions. The program therefore fails an exact deadline-renewal claim. The force-idle window condition and its completion error must be certified separately, as done below; the sequential probability formula is not assigned to this autonomous output.

9.3 The local field source during interaction

The free-output qualification in Corollary 26 is unnecessary for a specified smooth test under the same H_{tot}. Let $B(x)$ be a real symmetric smooth matrix, supported strictly inside the Dirichlet interval, with $\|B\|_\infty = G$; let $w \in C_c^\infty$ be real and supported in $[0, T]$. Use the same free-K vacuum subtraction and energy factors $\mathcal{Y} = (P, L_s Q)$, $L_s^* L_s = K$. Define

$$O_{\mathrm{int}} = \int w(t) \int [\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x)/2 - \text{free vacuum}_K]\, dx\, dt.$$

This is the material/carrier field source, not the controller's stress. By (17),

$$\mathcal{Y}(t) = \mathcal{Y}_0(t) + \Delta\mathcal{Y}(t), \qquad \Delta\mathcal{Y}(t) = \sum_a \int_0^t M_{t-s} h_a B_a(s)\, ds,$$

where $M_\tau h = (\cos(\omega\tau)h, L_s\omega^{-1}\sin(\omega\tau)h)$ and $\|M_\tau h\| = \|h\|$. As a vector of joint-state operators, $\|\Delta\mathcal{Y}(t)\psi\| \le W(t)\|\psi\|$, $W(t) = t\sum_a \|B_a\|\|h_a\|$.

Proposition 15 (An interacting-window field-source bound)**.** *Assume smooth coefficients of K and the smooth compact h_a above. For each a, define finite constants*

$$D_{P,a} = \sup_{0\le\tau\le T} \|\omega^2 [B M_\tau h_a]_P\|,$$
$$D_{S,a} = \sup_{0\le\tau\le T} \|\omega^2 L_s^* [B M_\tau h_a]_S\|.$$

With $\omega_i \ge c_- i$ as in Section A.3, set

$$C_0(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-3/2}\zeta(3/2) + D_{S,a} c_-^{-5/2}\zeta(5/2)],$$
$$C_1(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-2}\zeta(2) + D_{S,a} c_-^{-3}\zeta(3)].$$

For any $\Lambda > 0$, O_{int} is a Hermitian form with

$$\|(H_F + \Lambda)^{-1/2} O_{\mathrm{int}} (H_F + \Lambda)^{-1/2}\| \le C_{\mathrm{int}}, \tag{20}$$

where

$$C_{\mathrm{int}} = C_O + \int |w(t)| \left[\frac{C_1(t)}{\sqrt{\Lambda}} + \frac{C_0(t) + GW(t)^2/2}{\Lambda}\right] dt.$$

Here C_O is the free source constant with time test w. All references and the finite controller are retained. For two initial joint vectors of rms at most M_F and vector distance e, their actual interacting-window source difference is at most

$$2C_{\mathrm{int}}\sqrt{\Lambda e^2 + 2M_F e}\sqrt{\Lambda + M_F}. \tag{21}$$

Proof. The free part is Proposition 25. The response-square term is bounded by $GW(t)^2/2$ without assuming commuting response components. For the symmetrized cross term, let $v_i(t,x)$ be the free annihilation coefficient of mode i and $C_i(t) = \int v_i(t,x)^T B(x)\Delta\mathcal{Y}(t,x)\, dx$. Projection of the two response components onto e_i and insertion of ω^2 give

$$\|C_i(t)\| \le \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a}\omega_i^{-3/2} + D_{S,a}\omega_i^{-5/2}].$$

Interior multiplication by smooth B removes boundary compatibility conditions. Iterating the elliptic operator $K = -I\partial_x^2 +$ smooth lower-order terms controls the necessary Sobolev orders of propagated compact h_a, proving finiteness of the displayed graph norms uniformly over the compact time interval. Thus $\sum_i \|C_i(t)\| \le C_0(t)$ and $\sum_i \|C_i(t)\|/\sqrt{\omega_i} \le C_1(t)$.

For $E = \langle H_F\rangle_\psi$, $\|a_i\psi\| \le \sqrt{E/\omega_i}$ and $\|a_i^*\psi\| \le \sqrt{E/\omega_i} + 1$. Bounding $\langle a_i C_i\rangle$ by $\|a_i^*\psi\|\|C_i\psi\|$, and $\langle a_i^* C_i^*\rangle$ by $\|a_i\psi\|\|C_i^*\psi\|$, bounds the cross expectation by $2C_1(t)\sqrt{E}+C_0(t)$. No commutation between C_i and a_i has been used. The C_0 term is the creation contribution; treating the record as a c-number would incorrectly remove it. Apply $2\sqrt{E} \le (E+\Lambda)/\sqrt{\Lambda}$ and add the free and response bounds. Absolute convergence of the cross terms, the bounded response vector and the established free form give spectral-core convergence. Polarization proves (20). The weighted-form difference estimate of Corollary 26, with $\|H_F(\psi-\varphi)\| \le 2M_F$, proves (21). □

The constants do not depend on cursor site count if its coupling norms stay fixed. They do depend on the box, smooth test, profiles and time window. Rough spatial multipliers, boundary distributions and unsmeared stress are not covered. This result compares retained inputs under one common $H_{\rm tot}$; it does not compare arbitrary different apparatus Hamiltonians or supply the controller's missing stress.

9.4 A local contact energy and the interacting thermal reference

The inverse-K counterterm in (16) is useful for proof but is not a local physical density. For smooth compact material profiles $w_a(x)$ choose $h_a = (0, M(x)w_a(x))$ and replace it by

$$C_{\rm loc} = \frac{1}{2}\sum_{ab} B_a B_b \int w_a^T M w_b \, dx.$$

The actual mass-potential density becomes

$$V_{\rm loc}(x) = \tfrac{1}{2}[u - \sum_a B_a w_a]^T M [u - \sum_a B_a w_a]. \tag{22}$$

All gradient and current squares of the complete material remain. For each joint spectral branch $w = \sum_a b_a w_a$, the inverse-form variational formula gives

$$\langle (0,Mw), K^{-1}(0,Mw)\rangle = \sup_X \{2\langle w, Mu\rangle - q[X]\} \le \langle w, Mw\rangle,$$

because $q[X] \ge \langle u, Mu\rangle$. Consequently $C_{\rm loc} - C_{\rm min} \ge 0$ is a bounded controller operator. The actual local-contact Hamiltonian is $D^* H_F D + H'_D$ with $H'_D = H_D + C_{\rm loc} - C_{\rm min} \ge 0$. The positivity, rms and source proofs therefore apply to this physical contact, using its actual $C_{\rm loc}$ and the adjusted controller norm. No local stress is assigned to a nonlocal inverse operator.

Proposition 16 (Interacting equilibrium and retained record capacity)**.** *For either contact, put $d = \dim \mathcal{H}_D$, $d_0 = \|H'_D\|$, and $Z_F(\beta) = \operatorname{Tr} e^{-\beta H_F}$. Then the actual interacting Gibbs state exists and*

$$\begin{aligned} d e^{-\beta d_0} Z_F(\beta) \le Z_{\rm tot}(\beta) \le d Z_F(\beta), \\ \langle H_{\rm tot}^2\rangle_\beta \le \frac{16 e^{\beta d_0}}{e^2\beta^2} \frac{Z_F(\beta/2)}{Z_F(\beta)}. \end{aligned} \tag{23}$$

If independent uniform input bits $X_1, \dots, X_N$ are retained in the accessible controller and material with field mean energy at most M, and each is later readable with error $\epsilon_j \le 1/2$, then

$$\sum_j [\log 2 - h_2(\epsilon_j)] \le \log d + \inf_{\beta>0} [\beta M + \log Z_F(\beta)]. \tag{24}$$

Any decoder-accessible outside reference is an additional resource.

Proof. The eigenvalues of $D^* H_F D + H'_D$ lie between those of $D^* H_F D$ and those eigenvalues plus d_0, by min–max. Summing scalar exponentials proves the partition bounds; operator monotonicity of the exponential is neither available nor needed. For $x \ge 0$, $x^2 e^{-\beta x/2} \le 16/(e^2\beta^2)$. The partition ratio then proves the moment bound; its controller multiplicity cancels. Subadditivity and the field Gibbs variational inequality give $S(\rho) \le \log d + \beta M + \log Z_F(\beta)$. Binary Fano, conditional entropy subadditivity for the independent classical inputs, and data processing give $\sum_j [\log 2 - h_2(\epsilon_j)] \le I(X^N : \text{apparatus}) \le S(\rho)$. Take the infimum. The frequency bound also yields $\log Z_F(\beta) \le q/(1-q)^2$, $q = e^{-\beta c_-}$, so the capacity is finite for fixed resources. □

The same-H Gibbs purification is a graph-domain input, not the product Gaussian in (19). Stationary equilibrium supplies neither directed cursor nor blank records; $\beta^{-1} D(\rho_{\text{prep}} \| \rho_\beta)$ is a finite running state's free-energy excess, not its preparation protocol. At fixed energy, cursor dimension increases record capacity only logarithmically. Any outside history bank or entropy channel remains a resource.

10 Finite renewal and continuing return

10.1 Replacing a port is not renewing the interacting material

There are three distinct thermal operations below: replacement of a specified finite port; local thermal restriction on a free-carrier window; and preparation of the whole interacting Gibbs state. Only the first two have the stated finite operating protocols.

Proposition 17 (Universal held-content preservation forces a product extension)**.** *Let a channel $\mathcal{E}$ append a finite-dimensional port B to a held d-dimensional system A, whose marginal is the identity channel on every input, including all reference extensions. Then $\mathcal{E}(\rho) = \rho \otimes \tau$ for one fixed port state τ. Thus such a universal nondisturbing preparation cannot reproduce a correlated interacting Gibbs state. For $H = J Z_A Z_B$, $x = \beta J$, its Gibbs state is $\gamma = (I - \tanh(x) Z_A Z_B)/4$. At held input $I/2$ every such extension has half trace distance from γ at least $|\tanh x|/2$, attained by $\tau = I/2$.*

If instead only the classical Z_A label is conserved while conditional port Gibbs states $\tau_\pm = (I \mp \tanh(x) Z_B)/2$ are installed, the largest retained coherence factor is $\operatorname{sech} x$*. After the best held-system phase correction, an input $|+\rangle$ therefore suffers half trace distance at least $(1 - \operatorname{sech} x)/2$.*

Proof. Dilate the channel and purify its fixed ancilla. A pure held input must have a pure unchanged held marginal; hence the isometry sends $|\psi\rangle$ to $|\psi\rangle|e_\psi\rangle$. Applying this to basis

states and their superpositions forces one common environment vector. Tracing the environment gives the product extension. Equivalently the identity channel has rank-one Choi state. For the Gibbs example the observable $Z_A Z_B$ has norm one, expectation zero in $(I/2)\otimes\tau$ and $-\tanh x$ in γ, giving the distance bound; direct diagonal subtraction proves attainment. In a Z-nondemolition conditional preparation the two environment/port vectors purify τ_+, τ_-. Their overlap is at most their root fidelity, $\sum_z \sqrt{p_+(z)p_-(z)} = \operatorname{sech} x$, with equal Schmidt bases attaining it. That overlap multiplies the held off-diagonal matrix element. The qubit trace-distance formula proves the last assertion. □

This is a universal identity-channel condition, not a prohibition on preparing one specified state or maintaining a selected classical content. It supplies no universal constant for arbitrary approximate identity channels. An entire-body SWAP can prepare a fresh correlated copy, but it moves the old body's quantum content and correlations to the spent copy; it is not that identity channel.

For an attainable finite update, specify instead one receiver cell E with $H_E = \delta n_E$ and a fresh bank cell B with the same gap, Gibbs state η_β, and retained purifier. All held material, earlier records and old references form $\mathcal{O}$; the previous state of E may be correlated with all of them. The true S_{EB} commutes with $H_E + H_B$. The positive pulse $g(t)(I - S_{EB})$, $g \geq 0$, $\int g = \pi/2$, implements the true SWAP at finite duration, at least $\pi/(2g_{\max})$. At its endpoint E is η_β independently of the old $\mathcal{O}E$ state. That complete old state now belongs to $\mathcal{O}B$, with its correlations retained. The new cell remains correlated with its own Gibbs purifier. The $\mathcal{O}$ marginal is unchanged nonselectively; old records stored in E have moved to the spent slot rather than being erased. For an isolated prescribed pulse its switch work is zero because the bare cell energy is conserved and the pulse vanishes at the endpoints. This is not zero total controller or preparation cost.

Such a SWAP is also a bounded edge gate in H_D of the positive common material Hamiltonian. The cell must be force-idle while it is replaced. If n_E is still coupled to the material, $[S_{EB}, n_E Q(h)] \neq 0$; the matched gap alone gives no exact reset for that interacting contact. A finite-window estimate can retain this imperfection explicitly.

Proposition 18 (Finite-window port renewal with a live material)**.** *Compare the actual autonomous interface with an ideal renewal evolution having the same H_D, matched cell gaps and SWAP edge gate, but no material force in its renewal window $[0, T_R]$. Suppose their difference on the reference trajectory is $-\sum_a \Delta B_a Q(h_a) + \Delta C$, where $\Delta B_a = P_{\text{leak}} \Delta B_a$, $\Delta C = P_{\text{leak}} \Delta C$ and these controller operators commute with H_F. Here P_{leak} is a controller orthogonal projection; the differences are self-adjoint, so their displayed one-sided support is also two-sided support. Write $b_a = \|\Delta B_a\|$, $c = \|\Delta C\|$, $p(s) = \|P_{\text{leak}} \psi_s^i\|^2$, and assume $\|H_F \psi_s^i\| \leq M$. Define*

$$L_\Delta = \sqrt{2} \sum_a b_a \|\omega^{-1} h_a\|, \qquad c_\Delta = c + \sum_a b_a \|\omega^{-1/2} h_a\| / \sqrt{2}.$$

Including the bank and every old reference, the vector error is at most

$$\epsilon_R \leq \int_0^{T_R} [L_\Delta \sqrt{M}\, p(s)^{1/4} + c_\Delta \sqrt{p(s)}]\, ds. \tag{25}$$

Any specified incomplete ideal-SWAP or clock error is added separately. The same bound controls half trace distance of the renewed cell and held-reference marginal from their ideal outputs.

Proof. Duhamel bounds the vector difference by the integrated norm of the Hamiltonian difference on ψ_s^i. Apply the linear-field estimate to $P_{\text{leak}}\psi_s^i$. Its norm is $\sqrt{p}$ and

$$\|H_F^{1/2} P_{\text{leak}}\psi_s^i\|^2 \le \|P_{\text{leak}}\psi_s^i\|\|H_F P_{\text{leak}}\psi_s^i\| \le M\sqrt{p}.$$

The field and counterterm contributions give (25). Pure-vector distance bounds half trace distance, which contracts under the stated restrictions. □

Corollary 19 (A constructive finite receiver-renewal family)**.** *Fix $A > 0$, $0 < p < 1/2$, a matched receiver/bank pair and a material smearing h. For even m, use the positive spin-history clock with $J = A/m$, a true receiver/bank SWAP on the first edge and identities thereafter. Put $\chi_m = \mathbf{1}_{\{j \ge m/2\}}$ and*

$$H_m = H_{\text{hist}} + H_E + H_B + H_F - \chi_m n_E Q(h) + \tfrac{1}{2}\chi_m n_E \langle h, K^{-1}h\rangle.$$

Its initial cursor is $j = 0$; the material/old receiver state and all old references may be correlated, with initial physical rms $M < \infty$. The independent bank is Gibbs with its own purifier retained. This is a positive autonomous Hamiltonian. At $t_R = (m/A)\arcsin\sqrt{p}$ its joint half trace distance from a completed receiver renewal, with an explicitly specified comparison clock, is at most

$$2(1-p)^{m/2} + \frac{m \arcsin\sqrt{p}}{A}[L_\Delta \sqrt{M}\, e^{-mD/4} + c_\Delta e^{-mD/2}], \quad D = D_{\text{Ber}}(1/2\|p) > 0. \tag{26}$$

It tends to zero while the clock mean energy A, its norm bound $2A$ and the common-H physical rms certificate remain bounded independently of m. Clock space and duration grow; the bank is not cycled for free.

Proof. The field/contact terms are the positive controlled-displacement Hamiltonian of Proposition 14; the history clock and matched cell energies are nonnegative. For the force-free reference, each prefix is either the true SWAP or identity and commutes with $H_E + H_B + H_F$. Prefix conjugation therefore gives the same binomial cursor law for every correlated input. For $s \le t_R$ its binomial parameter is at most p; Chernoff gives $\Pr(j \ge m/2) \le e^{-mD}$. The ideal physical rms is the unchanged M. Proposition 18 proves the second term. The no-SWAP probability is $(1-p)^m$. As comparison use the same untrimmed force-free cursor with the completed SWAP and freely evolved material/reference state. Only its $j = 0$ branch differs from the ideal joint state; the retained-purification vector error is at most $2\sqrt{(1-p)^m}$. Half trace distance is no larger. No clock outcome is postselected. Triangle inequality proves (26). Finally $\|H_{\text{hist}}\| \le 2A$ and its initial mean is A. The cell norms and force/counterterm norms are fixed, so the initial total rms and the physical bound of Proposition 14 are uniform. □

This condition has a finite nonempty realization. Take the positive spin-history clock with $m = 64$, $J = .02$, the matched SWAP on its first edge, and a material force $-.2\,\mathbf{1}_{\{j\ge 32\}} n_E Q(h)$

with $\omega = 1$ and unit one-mode smearing. At $T_R = \arcsin\sqrt{.1}/J = 16.08753$ the force-free reference cursor has Bin$(64, .1)$ law; throughout this window its source tail is at most 7.04652×10^{-16}. Its uncompleted first-edge probability is $.9^{64} = .00117902$. For vacuum material input $M = 0$, (25) is below 6.9×10^{-8}. Completing the ideal SWAP on the same untrimmed cursor costs at most $2\sqrt{.9^{64}} = .068674$ in vector norm. The clock's mean energy is 1.28. Direct sparse evolution of the full force-on Hamiltonian at Fock cutoffs $6, 10, 14$, retaining a correlated held-body/receiver input, its old reference and the fresh bank purifier, gives vector difference 6.54×10^{-10} from the same force-free reference. The receiver's half trace distance from its bank state is .00043624; the held-body/reference marginal changes by less than 2.5×10^{-13} and energy conservation residual is below 4.1×10^{-13}. These are finite-regulator diagnostics of the nonempty renewal window, not a continuum proof. This is a paid receiver-renewal block with given correlated input, not an assertion that a previous program produces its cursor for free.

Actual leakage and completion require paid force-idle padding. The receiver certificate preserves held material and old content, but neither prepares its interacting Gibbs state nor implements an entropy outlet. A J-cell bank retains J spent slots, outputs and purifiers; each isolated replacement transfers excess energy $\delta(\langle n_E\rangle - \nu)$ of either sign. Preparation, clock, storage and removal remain charged. Q-RAIF B's oscillator-bank formulas [26] concern its different material. Repeated noisy readings of one command also do not justify $J \log 2$ in the independent-bit capacity bound (24).

10.2 Several rounds on one running material

The next reading can be installed on the same cursor that performs replacement. Fix a finite sequence of permitted data gates U_b, including receiver–bank swaps, and a nonnegative data energy H_0 with $[U_b, H_0] = 0$. Degenerate logic and matched-cell exchanges meet this condition. Keep the body, all banks, their purifiers, previous records and reversible workspace in the input. Between gates use the linear material forces and completing terms of Eq. (16).

Choose rational $0 < a_1 < \cdots < a_q < 1$ and integers m for which $k_b = ma_b$ are integral. Put U_b on edge k_b of the spin clock; other edges are identities. Each cursor interval has its own fixed force amplitudes, commuting within that interval. Orthogonal cursor supports make the full amplitudes commute. With J fixed, the resulting $H_m \geq 0$ is time independent and has common domain $\operatorname{Dom} H_F$. Its comparison schedule is

$$p(t) = \sin^2(Jt), \qquad t_b = J^{-1} \arcsin\sqrt{a_b}, \qquad T = \pi/(2J).$$

It applies the specified gates at t_b while the same complete material evolves throughout. Only the comparison uses these switching times; the actual apparatus has its gates and forces fixed in space along its cursor.

Proposition 20 (A finite sequence with its history retained)**.** *Let the smooth force profiles satisfy $h_\ell, \omega^{1/2} h_\ell \in L^2$, and fix an input purification with $\|H_F\psi\| \leq M_0$, including arbitrary old references. The actual H_m approximates the scheduled sequence jointly with its complete, untrimmed cursor, at every fixed checkpoint away from the t_b. At T all data gates have been included. The joint vector error tends to zero as $m \to \infty$, uniformly on this input class.*

Both actual and comparison material rms energies are bounded by a constant independent of m.

More explicitly, set

$$L_P = \sqrt{2}\sum_\ell \|B_\ell\|\|h_\ell\|, \qquad c_P = \frac{1}{\sqrt{2}}\sum_\ell \|B_\ell\|\|\omega^{1/2}h_\ell\|.$$

For any energy $E_* > 0$ a common rms bound is

$$M_T = \left[\sqrt{M_0 + E_*} + \frac{T}{2}\left(L_P + \frac{c_P}{\sqrt{E_*}}\right)\right]^2 - E_*. \tag{27}$$

In the prefix gauge, let H_b be the material/data Hamiltonian in interval b, and choose K_b so that $\|(H_b - H_{b-1})\zeta\| \le K_b$ for normalized vectors with $\|H_F\zeta\| \le M_T$. These constants are finite by the linear field bound. Then the gauge-state error satisfies

$$e_m(t) \le \sum_{b=1}^{q} K_b \int_0^t e^{-m[p(s)-a_b]^2}\, ds. \tag{28}$$

At a physical checkpoint add $2\sum_b e^{-m[p(t)-a_b]^2}$; this additional term is zero at T. For $0 < \alpha_b < \min(a_b, 1-a_b)$ and $v_b = 2J\sqrt{(a_b-\alpha_b)(1-a_b-\alpha_b)}$,

$$e_m(T) \le \sum_b K_b \left[\frac{\sqrt{\pi}}{v_b\sqrt{m}} + Te^{-m\alpha_b^2}\right]. \tag{29}$$

Proof. The prefix conjugation used for the history clock transforms its hopping term to $mJ + 2JJ_x$ and each force to the corresponding prefix-conjugated data amplitude. The transformed material Hamiltonian is piecewise constant in j. Compare it with the product of the untrimmed spin-clock state and the data/material sequence H_b. The clock distribution is $\mathrm{Bin}(m, p(t))$. Writing both piecewise Hamiltonians as sums of their jumps gives

$$H_m^{\mathrm{gauge}} - H_{\mathrm{cmp}}(t) = \sum_b (\mathbf{1}_{\{j\ge k_b\}} - \mathbf{1}_{\{t\ge t_b\}})(H_b - H_{b-1}).$$

The norm of each clock factor on the comparison state is the square root of its wrong-side probability. Hoeffding's bound is $e^{-m[p(t)-a_b]^2}$. Duhamel's formula therefore proves (28). On $|p - a_b| \le \alpha_b$ one has $p' \ge v_b$; integrate the Gaussian there and bound its complement by $Te^{-m\alpha_b^2}$ to obtain (29). Undoing the gauge at a checkpoint costs at most twice the square root of the probability of an incorrect prefix. At T the reference cursor is exactly $|m\rangle$, giving no prefix error.

For the energy estimate, all controller terms commute with H_F, even though their norms grow with m. On the energy core, $[H_F, Q(h)] = -iP(h)$ and

$$\left\|\sum_\ell B_\ell P(h_\ell)\zeta\right\| \le L_P\|H_F^{1/2}\zeta\| + c_P\|\zeta\|.$$

For $x(t) = \|H_F\psi_t\|$ this gives $D^+x \le L_P\sqrt{x} + c_P$. Hence $D^+\sqrt{x + E_*} \le (L_P + c_P/\sqrt{E_*})/2$, which proves (27). The data gates commute with H_F and cause no jump in this bound. Finite-mode energy-core regularization gives the same estimate before removing the cutoff; the common-domain propagator limit and closed-operator lower semicontinuity extend it to $\operatorname{Dom} H_F$. The linear field bound then supplies the K_b used above. Every estimate acts on the full purification and tensors with the old reference identity. □

For prescribed finite N, compile acquisition, decision, record force and bank swap on this path. Proposition 58 supplies one decision circuit from actual meter records and retained posterior weights; its repair flag can drive the force. Registers and thermal pairs are prepared initially, and spent correlations remain. At checkpoints, (21) and (27) certify a later reading and source test under the same subsequent H_m.

As a finite diagnostic, two probe readings of one oscillator, an intervening receiver–bank swap and a later force controlled by both records give endpoint joint vector errors .0701594, .0355814, .0192136, .0107206, .00393182 at $m = 16, 32, 64, 128, 512$, with $J = 1$. An old probe reference and the spent bank are retained. At $p = 7/16$, after the swap and before the second probe interval, receiver excitation is .0321012 for $m = 128$ and .00089552 for $m = 512$; endpoint accuracy alone would miss this timing error. The $m = 128$ final coordinate is .0393364 against the scheduled .0393304; removing the record-controlled forces gives -2.89×10^{-6} against zero. Fock cutoffs 8 and 10 change the endpoint vector error by 1.1×10^{-11}. These are regulator checks of the construction.

The clock uses $m + 1$ states, mean energy mJ, variance mJ^2, norm $2mJ$ and maximal edge coefficient at most $J(m+1)/2$. Fixed duration thus costs growing energy; fixed-energy renewal instead costs growing duration. Neither is indefinite fixed-capacity operation, and this material-source bound excludes controller covariant stress.

Appendix K gives a further use of this program: local linear-current pulses implement stress flows in the neutral $c = 1$ carrier, uniformly on bounded mean-energy inputs. The construction supplies a physical permission used in finite modular-geometry tasks, with explicit control and coherence conditions.

The comparison clock need not have the actual clock's energy. For one oscillator take $H_F = \omega a^\dagger a$, $Q = (a + a^\dagger)/\sqrt{2}$, $V = -bQ + C$ and $C = b^2/(2\omega)$. Use identity clock edges and turn on V for cursor sites $j \ge m/2$, with even m. The positive Hamiltonian is $H_m = H_{\mathrm{clock,m}} + H_F + \mathbf{1}_{j\ge m/2}V$. Starting in $|0\rangle_{\mathrm{clock}}|0\rangle_F$, its conserved total energy is mJ. The scheduled endpoint is $|m\rangle \otimes \zeta$ up to phase, where $\zeta = e^{-i(T/2)(H_F+V)}|0\rangle$ and $\langle H_F + V\rangle_\zeta = C$. Consequently

$$\langle H_m\rangle_{\mathrm{scheduled}} - \langle H_m\rangle_{\mathrm{actual}} = C \quad \text{for every even } m. \tag{30}$$

The joint-state error still tends to zero. The finite work is paid by the actual clock and missed by its unperturbed product comparison.

This defect can be repaired for the total-energy observable. Rotate the terminal spin toward negative S_x by an angle θ. Its clock energy is $mJ - mJ\sin\theta$, and its vector distance from $|m\rangle$ is at most $|\theta|\sqrt{m}/2$. If $\epsilon_m(\theta)$ is its probability below $m/2$, the product state's total-energy excess is $-mJ\sin\theta + C - \epsilon_m(\theta)\langle V\rangle_\zeta$. It equals C at zero and is negative at $\theta = \arcsin[2C/(mJ)]$ for sufficiently large m: the last probability is at most $e^{-3m/8}$ once $\theta \le \pi/6$. Continuity therefore supplies an exactly energy-matched comparison state with an additional $O(m^{-1/2})$ vector error. This is a physical comparison vector, not an extra gate

supplied by the original dynamics. Matching one energy expectation does not match an extended stress distribution. HAFF-D uses the mass-work identity of its common action to keep this debit in a finite source comparison [23].

10.3 Local thermal windows and a continuing diagnostic return

Port replacement is distinct from using a later region of a correlated field. Appendix I.1 executes a local thermal preparation with a paid finite autonomous clock. Appendix I.2 proves its record-selected same-field conditional error from vacuum mutual information and controls the source with second moments. The complete outside purification stays present; a spatially separate window is not declared globally fresh. These free-CAR certificates require a new thermal matching proof before use on the nonlinear interacting material.

Appendix J.1 gives a separate finite-horizon return in which the actual meter word, not an inaccessible fault label, selects repair of the held body and a thermal exchange. Appendix J.2 supplies an actual terminal record and replacement of its specified probe. The full input bank, posterior arithmetic, reversible garbage, clock energy and retained output history belong to that horizon. The actual mixed-port quantum-material update above does not inherit their different deadline or mean-support guarantees.

11 A common covariant ledger: controller, clock and support

The local potential (22) has potential stress $-g_{\mu\nu}V_{\mathrm{loc}}$ if its variables are Lorentz scalars. That term alone omits the mechanism which moves its profile and changes its amplitude. A common local action makes this omission explicit and offers a candidate completion at the classical action and regulated canonical levels.

Take dynamical scalars $X = (\phi_1, \phi_2, Z, \theta, y, \chi, r_1, \dots, r_d)$ on a $(1+1)$-dimensional background of signature $(-,+)$. Here Z is a support texture, χ a clock coordinate, and r_j memory/control coordinates. Specify the positive target coframe

$$\begin{aligned} e_{\phi_a} &= d\phi_a + \lambda_a \theta f(Z) dZ, & e_Z &= m_Z dZ, \\ e_\theta &= \sqrt{I}\, d\theta, & e_y &= \sqrt{J}\, dy, \\ e_\chi &= \sqrt{I_\chi}\, d\chi, & e_{r_j} &= \sqrt{I_j}\, dr_j. \end{aligned}$$

with all diagonal coefficients strictly positive, and put $G = \sum e \otimes e$. Let $b_a(\chi, r)$ be smooth bounded functions, $w_a(Z)$ smooth profiles compact in the texture value, and $M(Z)$ a smooth strictly positive material matrix. With $u = (\sqrt{I}\theta, \sqrt{J}y)$ and $v = u - \sum_a b_a(\chi, r) w_a(Z)$, choose

$$S = -\int \sqrt{-g}\, [\tfrac{1}{2} G_{AB}(X) g^{\mu\nu} \partial_\mu X^A \partial_\nu X^B + V(X)]\, d^2x, \tag{31}$$

$$V = V_Z(Z) + V_{\mathrm{mem}}(r) + V_{\mathrm{clock}}(\chi) + \tfrac{1}{2} v^T M(Z) v \geq 0. \tag{32}$$

The last square replaces the material mass potential rather than adding a second copy of it. Z, χ, r are actual fields in this candidate, not prescribed positions, waveforms or cursor

amplitudes. A prepared finite-energy texture can localize the contact; localization is not produced by writing $w_a(Z)$.

Proposition 21 (Common-action stress and clock reaction)**.** *For smooth finite-energy classical data of* (31)*, metric variation gives*

$$T_{\mu\nu} = G_{AB}\partial_\mu X^A \partial_\nu X^B - g_{\mu\nu}[\tfrac{1}{2} G_{AB}\partial X^A \cdot \partial X^B + V]. \tag{33}$$

The Euler–Lagrange equations are

$$\nabla_\mu (G_{AB}\partial^\mu X^B) - \tfrac{1}{2}(\partial_A G_{BC})\partial X^B \cdot \partial X^C - \partial_A V = 0. \tag{34}$$

They imply $\nabla^\mu T_{\mu\nu} = 0$ for the complete matter action. In specified Minkowski coordinates its canonical energy is

$$H = \int [\tfrac{1}{2}\Pi^T G^{-1}\Pi + \tfrac{1}{2} X'^T G X' + V]\, dx \geq 0.$$

At finite Fourier resolution, the following specified canonical ordering has a nonnegative Friedrichs realization on the supplied charge sectors and finite real coordinate space.

Proof. Vary the inverse metric with all scalar variables held fixed and use $T_{\mu\nu} = -2(\delta S/\delta g^{\mu\nu})/\sqrt{-g}$. Integration by parts in a field variation gives (34); contraction with $\partial_\nu X^A$ gives the stress divergence. In Minkowski coordinates $\Pi_A = G_{AB}\dot{X}^B$, so the Legendre transform has the displayed positive squares. The coframe is nonsingular and triangular. For a spatial Fourier regulator, let q denote its finite real coordinates and compact scalar zero modes. The reduced Lagrangian has positive kinetic matrix $G_K(q)$; put $A_K = G_K^{-1}$ and let $W_K(q) \geq 0$ be its integrated gradient and potential energy. Use Lebesgue measure on the real coordinates and the scalar zero-mode line bundles of Proposition 22, with their charge/winding direct sum. On smooth compactly supported sections choose the divergence-form kinetic ordering

$$\mathfrak{q}_K[\psi] = \tfrac{1}{2}\int (D_i\psi)^*(A_K)^{ij}(D_j\psi)\, dq + \int W_K|\psi|^2\, dq.$$

The D_i are coordinate derivatives with the stated zero-mode boundary conditions. Smooth locally elliptic coefficients make this densely defined nonnegative form closable; its Friedrichs realization is the specified operator. If the interaction acts only on retained low modes, the complete free carrier tail is added positively, as in the charged regulator. This choice of canonical measure and ordering is not a claim of regulator-independent covariant quantization. □

In particular,

$$I_\chi \Box\chi = \partial_\chi V = V'_{\text{clock}} - \sum_a (\partial_\chi b_a) w_a^T M v.$$

The amplitude change reacts on the clock. The memory equations have the corresponding $\partial_{r_j} b_a$ force, and the support equation includes derivatives of w_a, M and the target coframe. Dropping these terms returns an externally driven model. For a free massless clock initially

having $\partial_t \chi = \Omega$ over a proper-length interval ℓ, its initial kinetic energy is at least $I_\chi \Omega^2 \ell/2$. Smooth spatial tapers and contact reaction add costs and change that free trajectory. A locally timelike clock gradient, texture, memory preparation and boundary conditions are specified initial data; they are not guaranteed throughout an arbitrary run.

The action includes clock/support reaction; canonical positivity is only in the chosen coordinates and finite regulator. Curved-space quantum positivity, renormalized continuum stress and causal propagation after cutoff removal require further proofs. Matching (16) requires quantum acquisition and transported gates, the prepared texture/charge sectors, and a common finite-cursor/source estimate. The Gaussian K needs static supplied Z and controlled freezing or small perturbations of r, χ; dynamical cross terms and backreaction otherwise change H. That approximation needs an error bound, as does matching HAFF D's dilaton coordinates and full supported quantum state.

12 Discussion

The exact representation problem and the material return problem now have different tests. A discarded-port channel is checked by its multiplicative domain; a closed propagation symbol by its shell and coefficient relations. An aligned actuator can share the latter generators and charge their spatial form while still losing exact reduced control. The common-action and finite-return constructions ask whether the records, permissions and sources needed for these operations can be implemented together.

Acquisition is a physical change of the old history, not a supplied source label. The actual record drives the same complete material and produces another reading. Its connected fluctuations contribute to the source; dephasing can preserve an immediate field marginal while changing later joint control. Physical moments and smooth-source forms certify that return without replacing an unbounded stress by a trace-distance estimate.

Renewal depends on what is held. Universal identity-channel preservation cannot append a correlated interacting equilibrium. A specified receiver can nevertheless be replaced by a paid Gibbs bank, with spent correlations retained; the full ring can be reset and its mixed target reinstalled through four thermal outlets. Neutral compression, packet loading and spectral sorting supply a complete-output converter, not merely thermal mean energy. Isospectral axes need not retain a heat-source contrast after optimal sorting, and accurate direct reset alone leaves its purifier redundancy. Persistent sharp-exhaust gain remains a joint implementation question.

The clock distinctions in the introduction fix the resource claims: bounded-energy slow receiver renewal, fixed-duration energy-growing finite rounds, and a spatial work field with horizon-dependent inertia. The full joint-state estimate for the latter retains every old reference and charges finite clock work in the source comparison. Static frame/endpoints and dynamical support are still different problems. The neutral-current stress compiler gives another finite physical control permission, with coherence and precision budgets; it is not noise-protected control over every charged sector.

Foundation III supplies a complementary finite semiclassical mean-action scalar/JT return at common initial constraints: a learned frame protects old content, its record changes the material source, and a later X-basis reading of the old probe carries the geometric re-

Interface question	Established construction	Required input or remaining bridge
Exact operations	Multiplicative domain; propagation symbol; retained loops	Spectral law, access and quadratic-space conditions
Calibrated charging	State-independent work with actual generator alignment	Centered independent input; commuting drift; a measured command–momentum map
Actual quantum return	Acquisition, full material response and second record under one positive H	Prepared ports, allowed gates, physical transport and finite cursor
Local source	Same-H smooth interacting field bound; local positive contact potential	Smooth interior tests and moments; controller/support stress is separate
Finite maintenance	Receiver-bank SWAP; correlated-window and diagnostic protocols	Paid bank, storage and clock; not full interacting-material Gibbs renewal
Repeated body renewal	Full ring contraction, mixed-target preparation, neutral code, packet loading and finite-carrier sorting	Four thermal outlets, paid buffer, prescribed gates and energy-constrained capacity
Massive spatial body	Guarded local full-angle gate and weighted physical stress tests	Paid mass profile and state; actual outlet renewal and moving support remain
Field–clock evolution	Retained-state finite schedule with matrix Ward and delayed boundary	Static walls/frame; smooth input norms and horizon-dependent inertia
Geometric flow tasks	Local linear-current synthesis of neutral $c = 1$ stress gates	Circle recurrence, coherent control and growing precision budgets
Covariant apparatus	Common local action with dynamic clock and support reaction	Quantum gate matching, charged continuum and renormalized total stress

Table 1: The main interfaces and their distinct burdens. Auxiliary apparatus proofs are collected in the appendices.

sponse into the spent cell [50]. Its first-order signal arises after acquisition; the cursor and old purifications remain. Matching that run to the complete quantum material here requires a physical state/source bridge. HAFF D's rest-frame mean-stress proposition identifies one excess-radiation sector under fixed covariance and linear contact assumptions, not the entire acquired-record apparatus. Neither construction treats a spent port as a fresh bath.

A world model must retain the carrier and conserved total source; an interface must specify actual records and executable access; a continuing observer must maintain the content and resources those records serve. The common-action candidate connects these responsibilities at the classical and regulated canonical levels. Its quantum acquisition, charged continuum, moving supports and total renormalized stress still need matching. Neither exact multiplicativity nor finite repair derives experience or public geometric agreement. Conservation-law and multimode bounds [36, 38, 39] do not settle all these architectures at once. Split/modular inclusions and virtual subsystems [43, 44, 42] organize retained access; formation and maintenance of its permissions remain questions for T-DOME and Imprint.

13 Conclusion

A coupling represents an algebra exactly only under specific conditions. After a non-atomic dephasing port is discarded, its multiplicative domain is the coupling commutant. Retention can change that answer: closed oscillator loops return their port exactly, while closed propagation retains a faithful copy of the whole internal algebra. The single-shell condition supplies the spatial quadratic structure, and the supplied time and mass anticommutator supply its Lorentzian extension.

An actual alignment of propagation and actuator generators gives a conditional work–shell relation. It does not identify a positive cost with an indefinite metric, and it does not remove the reduced operation's multiplicative defect. This separates a testable physical compatibility condition from an assumed representation map.

The finite material construction makes the records, source and next action part of one retained quantum history. Receiver renewal and smooth-source accuracy have explicit clock, preparation and energy conditions. The further ring, massive-body and spatial-clock certificates extend what can be implemented, but their distinct layouts still need a common supported quantum realization. Foundation III's finite semiclassical return and HAFF-D's sustained renewal and joint sharp-exhaust questions retain their separate burdens. Exact representation is an interface condition; it does not by itself generate the world, a permanent apparatus or a subject.

Certificate guide: Q-RAIF C technical supplement

The representation bridge, physical mechanism and core results remain in this paper. The complete technical appendices are in `QRAIF_C_supplement.pdf`, *Retained Material, Work Clocks and Renewal Certificates: Technical Supplement to Q-RAIF C*. This is supplementary documentation, not a separate paper. Appendix letters and all existing result and equation numbers are retained. References below link to the supplement, which links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Complete charged material	Appendix A; Propositions 22, 25	Regulated carrier and smooth source tests with weighted tails.
Finite thermal-field return	Appendix B; Propositions 27, 28	The same field and old correlations are retained.
Ring-body renewal and conversion	Appendix C; Propositions 29, 30, 32	Paid outlets and buffers; finite-carrier assumptions.
Spatial work clock and massive gate	Appendix D; Propositions 35, 37	Finite joint-state control under the supplied walls and frame.
Conserved control and local outlet	Appendices E, F; Propositions 38, 41	Battery, timing, output window and outlet costs are retained.
Preparation, contact and recoil	Appendices G, H; Propositions 45, 54	The selected source is not its mean covariance; reaction is physical.
Actual acquired-record return	Appendices I, J; Propositions 57, 58	Actual programs, delayed readings and charged finite windows.
Neutral stress-flow control	Appendix K; Proposition 60	Local current synthesis with stated recurrence and precision resources.

These certificates concern distinct finite layouts. They do not establish one supported quantum apparatus with sustained renewal and jointly sharp informative exhaust; the remaining matching conditions are stated in the main discussion.

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